

A Survey of AI-Driven Demand Forecasting, Personalized Food Recommendation, Waste-Prevention Optimization, and Equitable Surplus Redistribution in Institutional Cafeteria Management

Abstract

Institutional cafeterias serving corporates, colleges, and schools typically operate under static, undifferentiated pricing and menu policies, despite serving customer segments with materially different price sensitivity, purchasing power, and taste preferences. This inefficiency, compounded by menus that do not adapt to seasonal or trend-driven demand shifts, food that is prepared without accurate demand or waste forecasting, and reliance on a single surplus-food recipient, results in lost profit, elevated food waste, and financial risk. This paper surveys 21 selected sources across four thoroughly evidenced research threads: (i) demand and sales forecasting for institutional and food-service settings, (ii) personalized food and menu recommendation systems, (iii) AI/ML-driven waste-prevention optimization, and (iv) equitable multi-recipient surplus food donation and distribution. We synthesize findings from machine learning, operations research, computational social choice, and applied food-systems literature, and find that while each of these four problems has been addressed independently — and the surplus-distribution problem in particular is addressed by a deep and varied literature — no existing system integrates all of them into a single segment-aware decision framework for institutional food service. We use this gap to motivate a proposed multi-layer system architecture, presented in Fig. 1, and explicitly flag which of its modules remain thinly evidenced by the sources reviewed here.

I. Introduction

Food service operations in corporate campuses, colleges, and schools are frequently managed using a single pricing sheet and a single menu-planning process, regardless of the underlying differences between these institutional segments. Corporate cafeterias typically serve a price-inelastic, premium-oriented population; colleges serve a trend-sensitive population responsive to novelty; and schools serve a highly price-sensitive, volume-driven population. Treating these as one undifferentiated market forfeits margin in high-willingness-to-pay segments while suppressing demand in price-sensitive ones.

Beyond pricing, three further inefficiencies compound the problem. First, what is prepared each day is rarely grounded in rigorous demand forecasting, leading to over-preparation and waste or under-preparation and lost sales. Second, menus and recommendations are rarely personalized to individual or segment-level taste, even though personalization is well established in adjacent food-recommendation research. Third, surplus food is conventionally routed to a single non-governmental organization (NGO) partner, creating volatility in donation flow and limiting the diversity of community partnerships, despite a mature body of fair-allocation research showing how to distribute surplus equitably across multiple recipients.

This survey reviews the state of research and applied practice relevant to each of these problems, drawing specifically on a curated set of 21 sources spanning demand forecasting, food recommendation, waste-prevention optimization, and surplus donation. It evaluates the extent to which existing systems address

these problems jointly, and positions the resulting research gap as the basis for the proposed Smart Cafeteria System architecture shown in Fig. 1. Because the current reading list does not include literature on segment-aware pricing, targeted promotional conversion, cost-recovery discount floors, or agentic multi-agent coordination, the Pricing and Promotion engines of the proposed architecture are presented as motivated extensions rather than as independently evidenced findings, and this distinction is maintained throughout the survey.

II. Related Work

A. Demand and Sales Forecasting in Institutional and Food-Service Settings

Forecasting demand for institutional food service has been studied using a broad range of statistical and machine-learning techniques. A case study of cafeteria demand forecasting in an institutional setting evaluated multiple machine-learning techniques, providing an empirical baseline for this problem [1]. A university refectory case study similarly framed demand forecasting for food production as a machine-learning problem, building on earlier work that incorporated student meal-reservation data as a predictive feature [2]. Complementary work combining machine learning and statistical analysis further supports the use of hybrid statistical/ML pipelines for food demand prediction [3]. Life-cycle-oriented work has connected demand forecasting directly to food-waste and environmental outcomes, showing that a bakery case study using ML-based demand forecasting reduced life-cycle environmental impacts associated with over-production [4]. A Bayesian approach to predicting food and beverage sales in staff canteens and restaurants offers a probabilistic alternative to point-forecast regression and tree-based models, useful for capturing forecast uncertainty where historical data is limited [5]. An ensemble approach combining statistical analysis with gradient-boosting algorithms (XGBoost, CatBoost) applied to restaurant-scale demand forecasting further incorporates store-specific contextual variables such as location and consumer-satisfaction drivers [6]. Collectively, this cluster establishes that demand forecasting for institutional and food-service cafeterias is a mature, actively studied problem addressable through methods ranging from Bayesian regression to gradient-boosted ensembles, though none of these six studies incorporate segment-level differentiation by institution type — the specific framing targeted by the proposed AI Prediction Module's segment-prediction function.

B. Food and Menu Recommendation Systems

A second, related thread addresses recommending food items or menus to individual users rather than forecasting aggregate demand. A hybrid recommender system for an academic dining environment combined collaborative-filtering, content-based, and knowledge-based models, finding that an AdaBoost-based hybrid model outperformed decision-tree and k-nearest-neighbor baselines on a dataset of over 2,500 students [7]. MenuAI applies a transformer-based deep-learning architecture to restaurant menu recommendation, illustrating that sequence-model architectures established for language tasks transfer to food recommendation [8]. At the ingredient level, a convolutional-neural-network-based ingredient-detection pipeline paired with a recipe-matching algorithm demonstrates that visual recognition of available ingredients can drive recipe recommendation, reporting 94% ingredient-classification accuracy on a custom dataset [9]. This cluster is oriented toward what an individual is likely to want rather than

how a menu or price should adapt to which institutional segment, which motivates treating recommendation as informing, but not substituting for, the segment- and trend-aware logic in the proposed AI Prediction Module.

C. AI/ML-Driven Waste-Prevention Optimization

A third thread targets food waste reduction directly, largely by optimizing what is ordered, prepared, or offered before it becomes surplus. A fair-allocation formulation of menu selection frames the problem as choosing a menu across agents with heterogeneous dietary needs and serving sizes so as to minimize wastage under optimistic and pessimistic consumption models, applying computational social-choice techniques to a food-service ordering problem [10]. Empirical work using artificial intelligence to reduce food waste across restaurant operations quantifies the extent to which predictive and inventory-aware interventions lower discarded food volumes [11]. Complementary studies using machine-learning models for waste-reduction-oriented forecasting and optimization [12], and analyses of predictive models specifically for food-waste reduction [13], extend this cluster, reinforcing that waste is best addressed jointly through improved forecasting and optimization-driven menu or order-quantity decisions rather than through downstream disposal management alone. A broader survey of AI in modern consumer-driven food systems situates this cluster within the wider landscape of AI-for-food-systems applications, tracing uses from production through consumption and including subsidy-based university dining work that models over-preparation under demand uncertainty using artificial neural networks — one of the few threads targeting institutional, rather than purely commercial, food service specifically for waste prevention [14]. Unlike the food-bank literature discussed in Section II-D, this cluster targets waste prevention on the supply side, before food becomes surplus, and directly grounds the Waste Prediction module of the proposed architecture.

D. Equitable Multi-Recipient Surplus Food Donation and Distribution

The final and most extensively represented thread in this reading list addresses what should happen to food once it becomes surplus. Robust-optimization work modeling food-donation management under supply and demand uncertainty, developed in the COVID-19 context, formalizes donation planning as a stochastic optimization problem under volatile inflow and need [15]. A sustainability-assessment study of surplus food donation models the donation pathway as a transfer system and quantifies its environmental, economic, and social value, providing a framework for evaluating redistribution outcomes beyond simple waste-diversion volume [16]. Applied work describing a mobile app for surplus food donation illustrates the operational and technology-platform side of connecting surplus-generating institutions to recipient organizations [17]. A game-theoretic framework for surplus food distribution in smart cities models the interaction between multiple donating and receiving parties as a strategic game, offering an alternative to the optimization-based allocation formulations used elsewhere in this cluster [18]. Robust-optimization approaches for equitable and effective donated-food distribution formalize the allocation problem along effectiveness, efficiency, and equity dimensions using a dynamic-programming formulation validated on real food-bank data [19]. Forecasting has also been integrated directly into allocation: a hierarchical methodology predicts donation volumes across a multi-warehouse network and feeds these forecasts into an optimization model that minimizes the deviation between over-served and

under-served locations [20]. Vehicle-routing extensions of the same problem jointly optimize delivery efficiency alongside equitable and effective distribution among partner agencies [21]. Collectively, this seven-paper cluster provides a strong and varied methodological foundation — spanning stochastic optimization, sustainability accounting, applied platforms, game theory, dynamic programming, and vehicle routing — for the NGO Distribution module, though it centers on food banks and city-scale donation networks serving many recipients rather than institutional cafeterias distributing surplus to a small, relationship-tracked set of NGO partners — the smaller-scale, relationship-score-weighted variant targeted by the proposed system.

III. Comparative Analysis and Research Gap

Table I compares the four literature clusters identified in this survey against the capabilities central to the proposed system. No single cluster addresses more than two of the four capability columns, and — consistent with the note in Section II-E — none of the reviewed clusters provides any coverage of pricing, promotion, or agentic-coordination capabilities.

System / Study Cluster	Demand Forecasting	Personalized Recommendation	Waste-Prevention Optimization	Multi-NGO Split / Pricing / Agentic
Demand-forecasting studies [1]–[6]	Yes	No	Partial (order sizing)	No
Food & menu recommendation systems [7]–[9]	No	Yes	No	No
Waste-prevention optimization & AI-food-systems work [10]–[14]	Partial	No	Yes	No
Surplus-food donation & fair-allocation models [15]–[21]	Partial (donation volume, [20])	No	No	Yes (NGO split only)
Proposed System (Fig. 1)	Yes	Yes	Yes	Yes (pricing/promotion not yet evidenced)

The comparison indicates that demand forecasting, personalized recommendation, waste-prevention optimization, and equitable multi-NGO distribution are each independently mature and well evidenced by the current reading list, with surplus distribution in particular supported by a deep and methodologically varied literature. Institutional food service has not, however, been the setting in which these four capabilities are integrated, and no reviewed source addresses segment-tiered pricing, an enforced cost-recovery floor, or agentic coordination across these functions. This is the specific gap the proposed system architecture is intended to address, with the caveat that its Pricing and Promotion engines currently rest on a thinner evidentiary base than the other four intelligence-layer modules.

IV. Proposed System Architecture

Building on the gap identified above, Fig. 1 presents the proposed Smart Cafeteria System architecture. The architecture is organized into three layers.

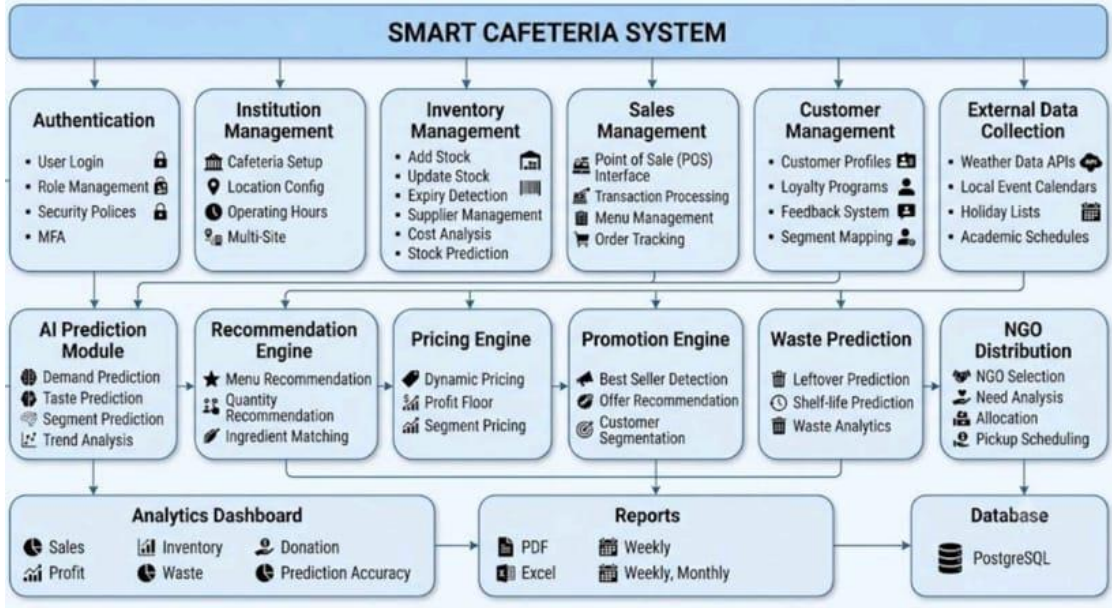


Fig. 1. Proposed Smart Cafeteria System architecture: operational/data layer, six-engine intelligence layer, and output layer.

The Operational and Data Layer comprises six modules that supply the raw operational and contextual data consumed by the layer above: Authentication (user login, role management, security policies, multi-factor authentication); Institution Management (cafeteria setup, location configuration, operating hours, multi-site support); Inventory Management (stock addition and updates, expiry detection, supplier management, cost analysis, stock prediction); Sales Management (point-of-sale interface, transaction processing, menu management, order tracking); Customer Management (customer profiles, loyalty programs, feedback system, segment mapping); and External Data Collection (weather data APIs, local event calendars, holiday lists, academic schedules).

The Intelligence Layer comprises six coordinating engines. The AI Prediction Module (demand prediction, taste prediction, segment prediction, trend analysis) is grounded in the demand- and sales-forecasting literature reviewed in Section II-A [1]–[6]. The Recommendation Engine (menu recommendation, quantity recommendation, ingredient matching) is grounded in the food- and menu-recommendation literature reviewed in Section II-B [7]–[9]. The Waste Prediction module (leftover prediction, shelf-life prediction, waste analytics) is grounded in the waste-prevention optimization literature reviewed in Section II-C [10]–[14]. The NGO Distribution module (NGO selection, need analysis, allocation, pickup scheduling) is grounded in the surplus-donation and fair-allocation literature reviewed in Section II-D [15]–[21]. The Pricing Engine (dynamic pricing, profit floor, segment pricing) and the Promotion Engine (best-seller detection, offer recommendation, customer segmentation) correspond to the segment-aware pricing, cost-recovery price-floor, and trial-conversion logic described in Section I; consistent with Section II-E,

these two engines remain the least evidenced by the current reading list and should be treated as design extensions pending dedicated pricing- and promotion-specific literature.

The Output Layer consists of an Analytics Dashboard (sales, profit, inventory, waste, donation, and prediction-accuracy views), a Reports module (PDF and Excel exports on weekly and monthly cadences), and a PostgreSQL database that persists operational, prediction, and transaction records generated across all upstream modules.

V. Open Challenges and Future Directions

- Evidentiary gap on pricing and promotion: as noted in Section II-E, segment-aware pricing, targeted conversion, cost-recovery discounting, and agentic coordination are not covered by the current reading list. Closing this gap with dedicated sources on dynamic retail pricing and multi-agent coordination is a prerequisite for defending the Pricing and Promotion engines with the same rigor as the rest of the architecture.
- Data availability: segment classification and taste-trend prediction require historical institutional sales data that is unlikely to exist for a new pilot deployment, motivating either synthetic data generation or an early partnership with an operating cafeteria vendor.
- Trend-source selection: the AI Prediction Module's trend-analysis function needs a concretely specified live-trend source (e.g., search-trend APIs or food-delivery bestseller feeds); this remains the least-grounded prediction function even setting aside the pricing/promotion gap, and needs its own feasibility study.
- Evaluation methodology: waste-reduction and profit-lift claims require a defined baseline and metric (e.g., percentage waste reduction, incremental margin versus flat pricing, conversion lift versus blanket discounting), ideally validated via historical backtesting or a controlled pilot.
- Identity tracking for conversion targeting: the Promotion Engine's targeting of non-buyers implies some form of per-customer identification (loyalty card, campus ID, or app login), which is an infrastructure dependency that should be stated as an explicit assumption.
- Fairness and transparency in segment pricing: differentiated pricing by institution type carries a perceived-fairness risk that should be addressed through transparent communication of the pricing basis, once pricing-specific literature is incorporated.
- Elasticity-aware discounting, automated segment auto-classification for new institutional clients, and NGO need-prediction (rather than reliance on NGO self-reporting) are natural extensions once the core engines are validated.

VI. Conclusion

This survey has examined 21 selected sources across four research threads relevant to institutional cafeteria management — demand and sales forecasting, personalized food and menu recommendation, AI/ML-driven waste-prevention optimization, and equitable multi-recipient surplus distribution — and found each to be independently well studied, with surplus distribution supported by a particularly deep and methodologically varied literature. None of the reviewed sources integrate these threads within a single institutional food-service system, and none address segment-tiered pricing, an enforced cost-recovery floor, or agentic coordination across functions. Framing the proposed system architecture (Fig. 1) around this gap offers a technically grounded design direction and a defensible novelty claim for its AI Prediction, Recommendation, Waste Prediction, and NGO Distribution engines; its Pricing and Promotion engines, by contrast, are explicitly flagged as extensions requiring further literature support before the novelty claim can be made with equal confidence across the full architecture.

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